

DBMS PROJECT

HOSPITAL MANAGEMENT SYSTEM

GROUP:

K. NAGENDRA	22CS01026
R. KARTHEEK	22CS01012
M. PARDHIVA SRI SAI	22CS01003
V. REDDY ADITHYA ROYAL	22CS01028

Demo Video : [Demo Video Link](#)

About the Project: Hospital Management System

The Hospital Management System (HMS) is a comprehensive software solution developed to simplify and automate the day-to-day operations of a hospital or healthcare center. It provides a centralized platform to manage patients, doctors, staff, appointments, medical records, billing, and inventory, all in one place.

This system addresses the challenges faced by hospitals in handling a large volume of data manually and ensures accuracy, speed, and efficiency in operations. By digitizing hospital workflows, HMS helps in reducing administrative burden, minimizing human errors, and enhancing the quality of patient care.

The project is designed with a user-friendly interface and role-based access, ensuring that different users (like doctors, nurses, receptionists, and administrators) can perform their tasks efficiently based on their responsibilities.

TECH STACK:

Frontend : React.js

Backend : Node.js + Express

Database : MySQL

TABLES :

Users:

- User_id (primary key)
- Name
- Email
- Password
- phone_no
- role ENUM('Doctor', 'Admin', 'FrontEntry', 'Nurse')

Doctors:

- d_id (primary key)
- User_id (FOREIGN KEY REFERENCES users(user_id))
- Specialisation
- Degree
- experience

Patients:

- p_id (primary key)
- name
- age
- gender
- bloodgroup (ENUM('A+', 'A-', 'B+', 'B-', 'AB+', 'AB-', 'O+', 'O-'))
- phoneno
- address
- email
- createdAt (TIMESTAMP)

Rooms:

- r_id (primary key)
- status (ENUM('Available', 'Occupied'))

Appointments:

- app_id (primary key)
- p_id (FOREIGN KEY REFERENCES patients(p_id))
- d_id (FOREIGN KEY REFERENCES doctors(d_id))
- a_date
- priority(Enum('Scheduled', 'Completed', 'Cancelled'))
- symptoms

Admissions:

- adm_id(primary_key)
- p_id (FOREIGN KEY REFERENCES patients(p_id))
- d_id (FOREIGN KEY REFERENCES doctors(d_id))
- r_id (FOREIGN KEY REFERENCES rooms(r_id))

Tests:

- t_id(primary key)
- test_name

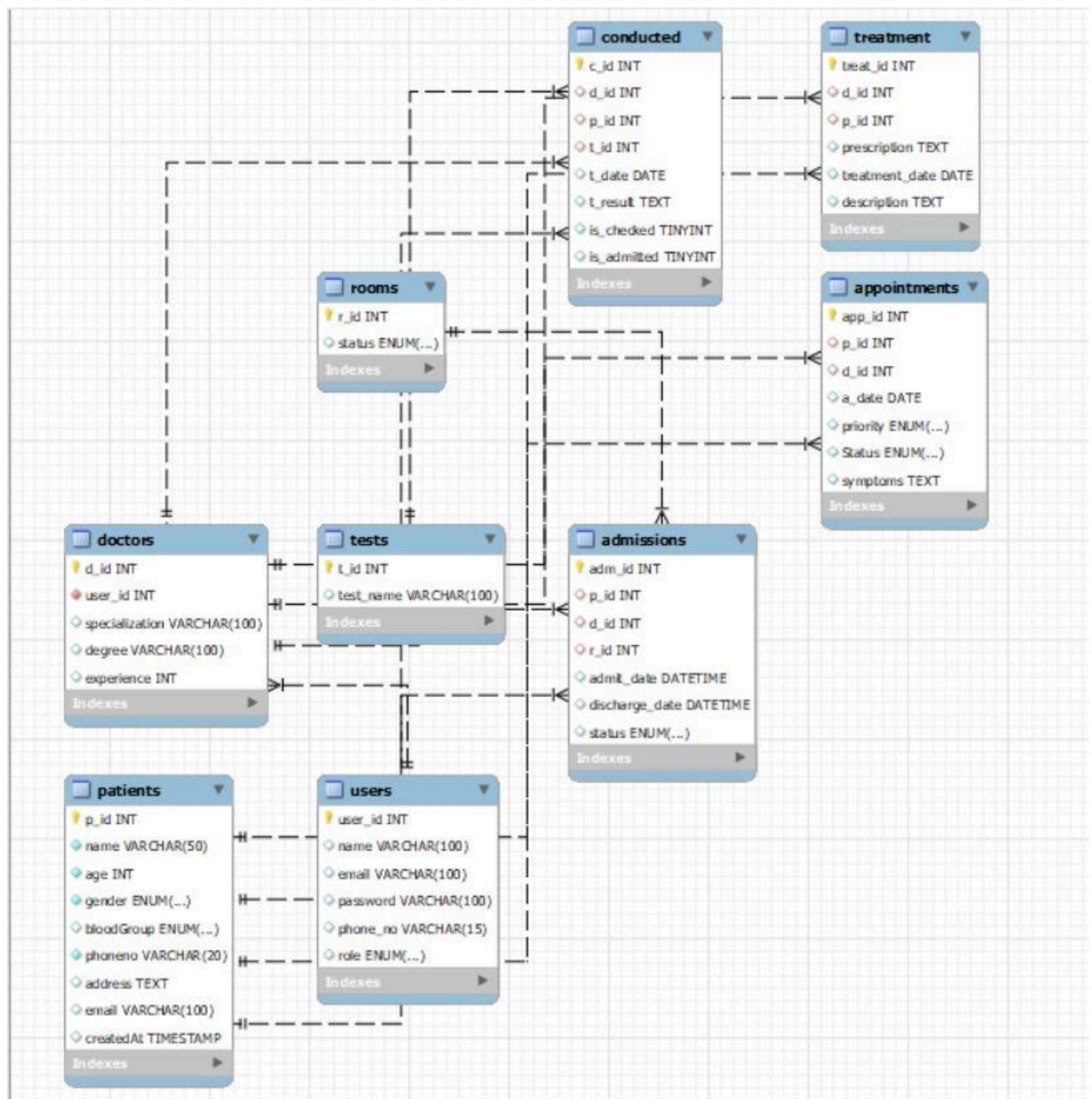
Conducted:

- c_id(primary key)
- p_id (FOREIGN KEY REFERENCES patients(p_id))
- d_id (FOREIGN KEY REFERENCES doctors(d_id))
- t_id (FOREIGN KEY (t_id) REFERENCES tests(t_id))
- t_date
- t_result
- is_checked
- is_admitted

Treatment:

- treat_id(primary key)
- p_id (FOREIGN KEY REFERENCES patients(p_id))
- d_id (FOREIGN KEY REFERENCES doctors(d_id))
- prescription
- description
- treatment_date

Relations:



USERS:

- ADMIN
- DOCTOR
- FRONT DESK OPERATOR
- NURSE (DATA ENTRY OPERATOR)

ADMIN:

Route: /admin

Role :

- Manages all users.
- Management involves adding, deleting, and updating users.
- Also creates a profile for each user.

DOCTOR:

Route: /doctor

Role :

- Treats Patients, Records treatments and Tests prescribed for them
- Add tests to be conducted

FRONT DESK OPERATOR:

Route: /frontend

Role:

- Registers patients, books appointments, admits/discharges patients, and checks room availability.

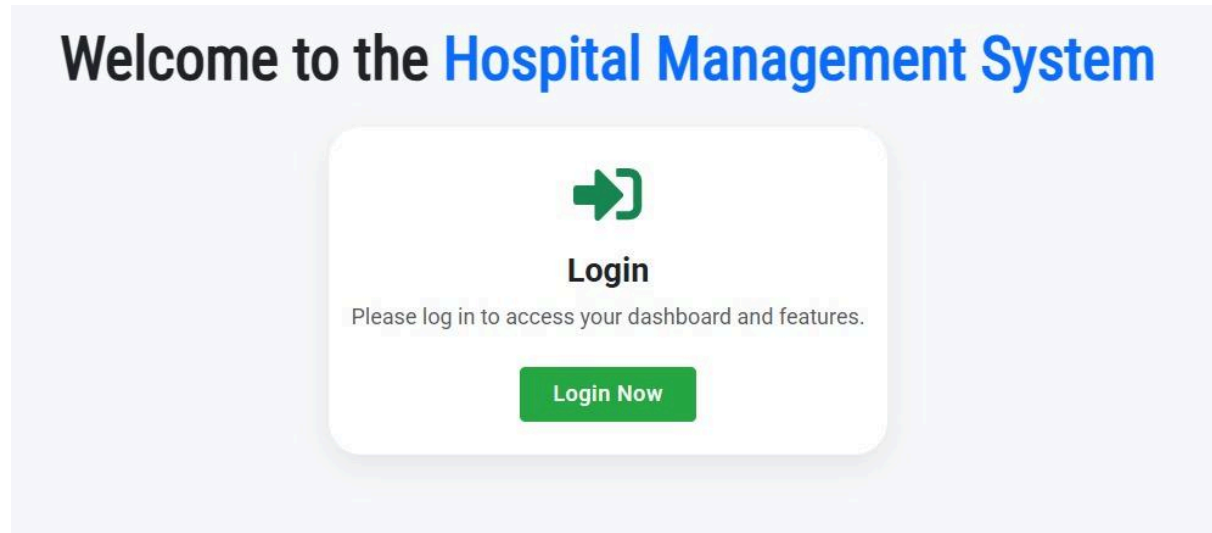
NURSE: (DATA ENTRY OPERATOR)

Route: /nurse

Role:

- Conducts tests for Patients, records those Results

WORKFLOW:



The system begins with a **login interface** where users must authenticate using their credentials. Upon successful login, users are redirected to their respective dashboards based on their assigned **role**, which can be one of the following:

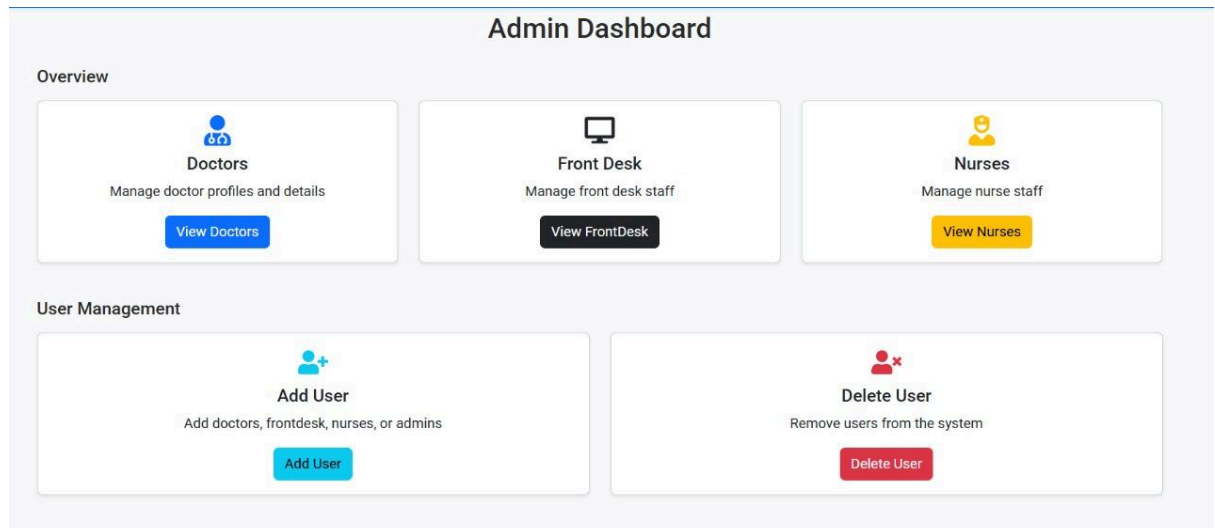
- **Administrator**
- **Doctor**
- **Front Desk Operator**
- **Nurse (Data Entry Operator)**

Each role provides access to specific functionalities tailored to the responsibilities of the user. This role-based access ensures secure and efficient management of hospital operations.

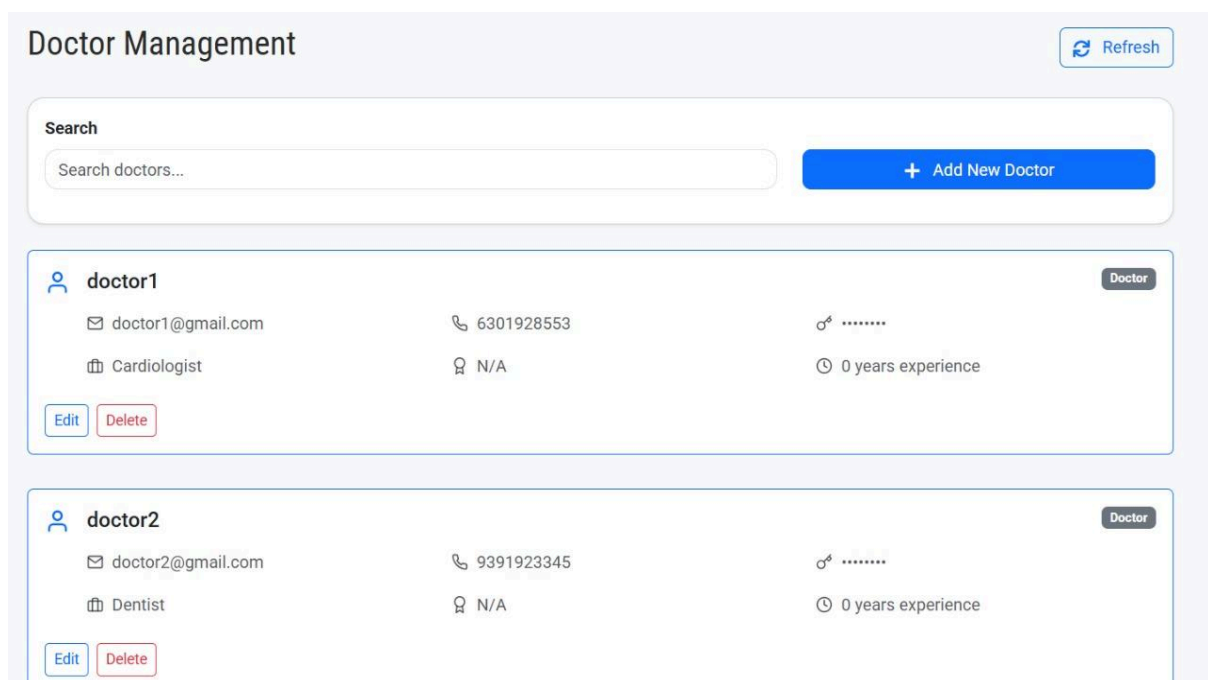
A screenshot of a "Hospital Login" form. The title "Hospital Login" is at the top in a large, bold font. Below the title is the text "Login as:" followed by a dropdown menu currently showing "Doctor". Below the dropdown is an "Email" input field containing "doctor@gmail.com". Below the email field is a "Password" input field with a masked password "....." and an eye icon to toggle visibility. At the bottom of the form is a blue button with the text "Login" in white.

Upon successful authentication, the user is directed to their respective dashboard based on their role. The workflow for each role is outlined below:

ADMIN:

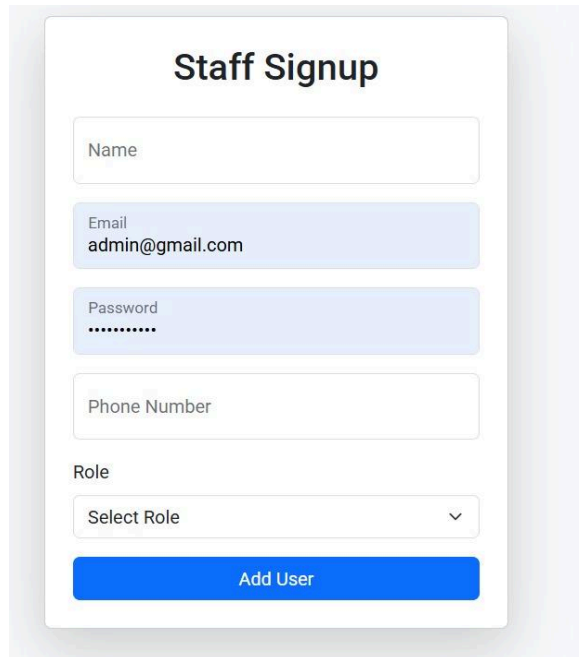


Upon clicking on the "View Doctors" option, the system displays a list of Doctors



Similarly, upon clicking on the "View Front Desk Operators", it will display all the front desk operators, and the same goes with the "View Nurses" Option

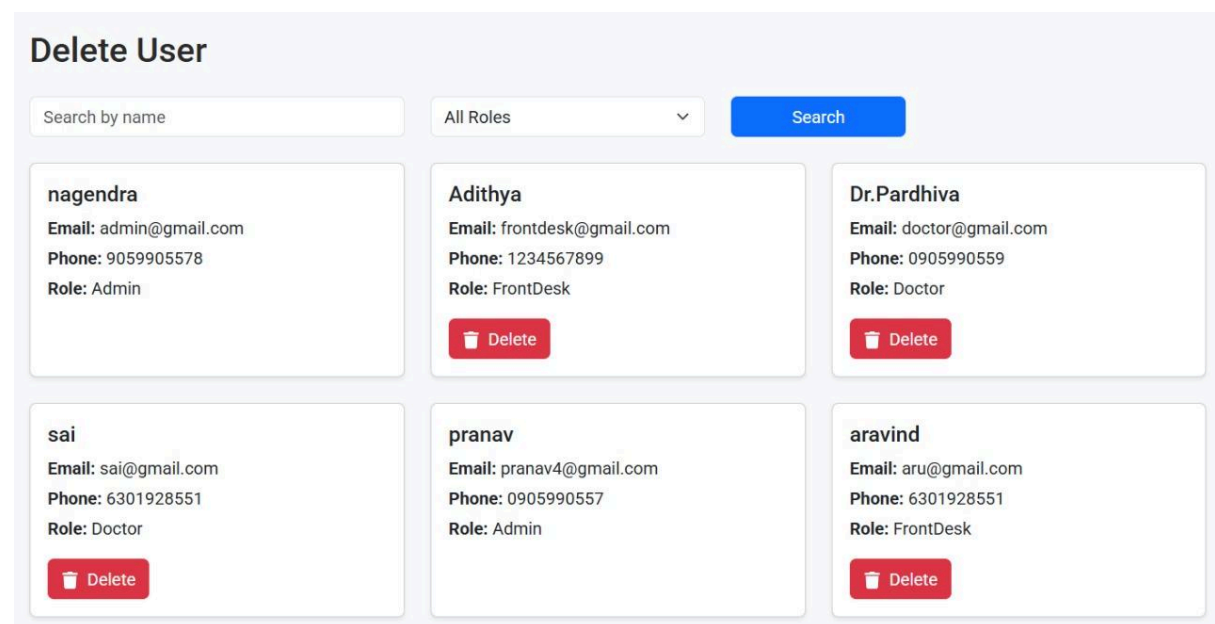
The Administrator can add a new user to the system by clicking on the “Add New User” option, which opens a form to enter user details such as name, role, email, and contact information.



A vertical form titled "Staff Signup" with a light blue background. It contains several input fields: "Name" (white), "Email" (light blue, containing "admin@gmail.com"), "Password" (light blue, containing "*****"), "Phone Number" (white), and "Role" (white dropdown menu with "Select Role" and a downward arrow). At the bottom is a blue "Add User" button.

The Administrator can remove an existing user from the system by selecting the “Delete User” option. This action allows the Admin to view a list of users and delete any user as required, ensuring secure and up-to-date user management.

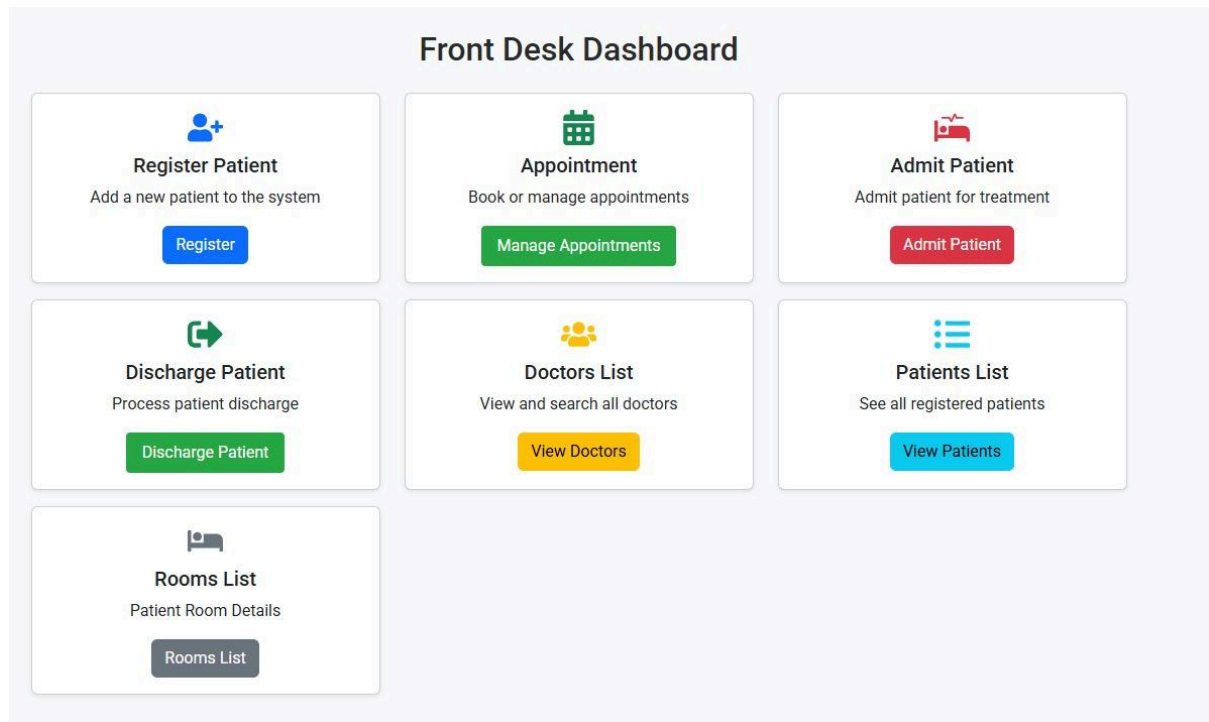
Note: *For security and integrity purposes, the system does not allow the Administrator to delete their account.*



An interface titled "Delete User" with a light blue background. It features a search bar labeled "Search by name", a dropdown menu labeled "All Roles", and a blue "Search" button. Below these are six user cards arranged in a 2x3 grid. Each card displays the user's name, email, phone number, role, and a red "Delete" button with a trash icon.

Name	Email	Phone	Role	Action
nagendra	admin@gmail.com	9059905578	Admin	Delete
Adithya	frontdesk@gmail.com	1234567899	FrontDesk	Delete
Dr.Pardhiva	doctor@gmail.com	0905990559	Doctor	Delete
sai	sai@gmail.com	6301928551	Doctor	Delete
pranav	pranav4@gmail.com	0905990557	Admin	
aravind	aru@gmail.com	6301928551	FrontDesk	Delete

Front Desk Operator:



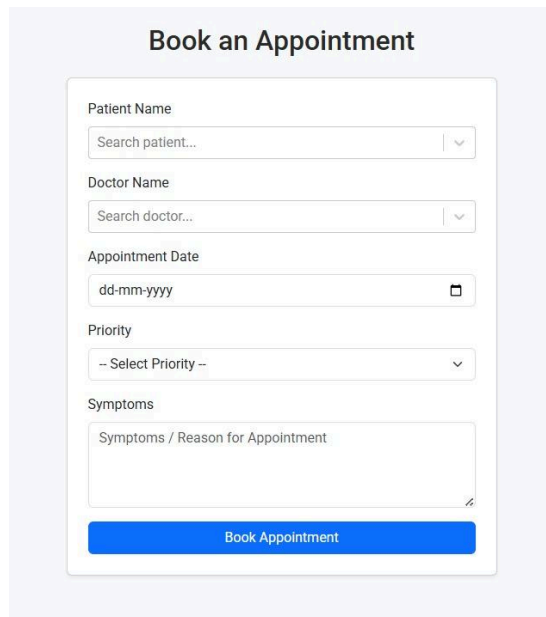
1. Register Patient:

The Front Desk Operator can register a new patient by selecting the “**Register**” option. This opens a registration form where essential patient details such as name, age, gender, contact information, and medical history can be entered and submitted to the system.

The image shows the 'Patient Registration' form. It contains input fields for Name, Age, Address, Phone, and Email. Below these are dropdown menus for Gender and Blood Group, both with '-- Select --' as the placeholder text. A blue 'Register Patient' button is at the bottom of the form.

2. Book Appointment:

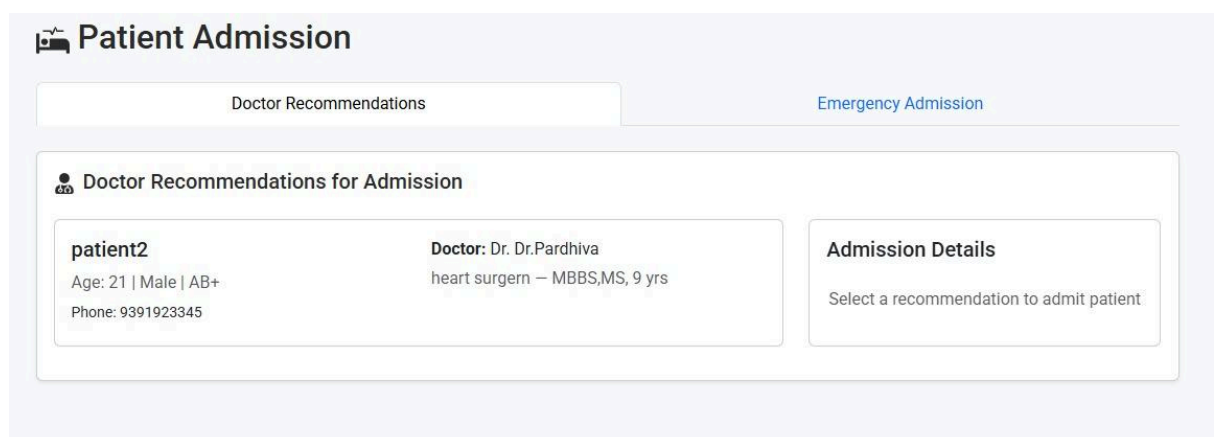
The Front Desk Operator can schedule an appointment by selecting the **“Manage Appointments”** option. This allows them to choose a doctor, specify the patient details, select a preferred date and time, set priority, and record symptoms before confirming the booking.



The 'Book an Appointment' form is a vertical stack of input fields. It starts with a 'Patient Name' field containing a search bar and a dropdown arrow. Below it is a 'Doctor Name' field with a similar search bar and dropdown. The 'Appointment Date' field shows a date picker with 'dd-mm-yyyy' and a calendar icon. The 'Priority' field is a dropdown menu currently showing '-- Select Priority --'. The 'Symptoms' field is a large text area with the placeholder 'Symptoms / Reason for Appointment'. At the bottom is a blue 'Book Appointment' button.

3. Admit Patient to Room:

Upon receiving the doctor's recommendation for admission, the Front Desk Operator can assign an available room to the patient by selecting the **“Admit Room”** option. The system displays a list of available rooms, allowing the operator to allocate one based on availability and patient requirements.



The 'Patient Admission' interface features a header with a medical icon and the title. Below the header are two tabs: 'Doctor Recommendations' (active) and 'Emergency Admission'. The main content area is titled 'Doctor Recommendations for Admission' and contains two panels. The left panel, titled 'patient2', displays patient information: 'Age: 21 | Male | AB+' and 'Phone: 9391923345'. The right panel, titled 'Admission Details', contains the text 'Select a recommendation to admit patient'. The doctor's recommendation, 'Doctor: Dr. Dr.Pardhiva heart surgern — MBBS,MS, 9 yrs', is displayed between the two panels.

4. Discharge Patient:

After the completion of treatment, the Front Desk Operator can discharge the patient by selecting the “**Discharge Patient**” option. This action updates the patient's status to discharge, records the discharge date, and marks the assigned room as available for future admissions.

Discharge Patients						
Admission ID	Patient Name	Age	Room	Admitted On	Doctor	Action
12	kartheek	89	4	19/04/2025	Dr.Pardhiva	Discharge

5. View Doctors / View Patients / View Rooms:

The “**View Doctors**”, “**View Patients**”, and “**View Rooms**” options allow the Front Desk Operator to access comprehensive lists of all doctors, patients, and hospital rooms.

- The **Doctors List** includes details such as name, specialisation, contact information, and availability.

Doctor List						
Search doctors by name...						
#	Name	Email	Phone	Specialization	Degree	Experience (yrs)
1	Dr.Pardhiva	doctor@gmail.com	0905990559	heart surgern	MBBS,MS	9
2	sai	sai@gmail.com	6301928551	ENT	MTech	2
3	doctor1	doctor1@gmail.com	6301928553	Cardiologist		
4	doctor2	doctor2@gmail.com	9391923345	Dentist		
5	pramod	pramod@gmail.com	6301928552	dogs veterinary		

- The **Patients List** displays patient ID and current admission or appointment status.

Patient List								
Search by name, email, phone, or blood group								
ID	Name	Age	Gender	Email	Phone	Blood Group	Address	Registered On
1	kartheek	89	Male	22CS01026@iitbbs.ac.in	0905990557	O+	pedamiram,west godavari	4/12/2025
2	pardhiva	18	Female	22CS01026@iitbbs.ac.in	0905990557	A+	pedamiram,west godavari	4/13/2025
3	patient1	20	Male	redrowthukartheek@gmail.com	9391923345	AB+	Avanigadda	4/16/2025
4	patient2	21	Male	abc@gmail.com	9391923345	AB+	avg	4/16/2025
5	patient3	22	Male	abcd@gmail.com	9391923345	A+	avg	4/16/2025

- The **Rooms List** shows room numbers along with their current status (e.g., Occupied, Available).

Room Assignments				
Room Number	Patient ID	Patient Name	Age	Doctor Name
4	1	kartheek	89	Dr.Pardhiva

Doctor:

Doctor Dashboard

Patient List

View and search patients under your care

View Patients

Inbox

See test results and emergency alerts

Open Inbox

Test Results

Review tests and assign rooms

Test Results Room Assignment

Add Test

Add new tests

Add Test

1. Patient List:

The Doctor can view a comprehensive list of all patients they have treated by selecting the “**View Patient List**” option. This list includes essential patient details such as name, age, gender, medical history, and appointment or admission records associated with the doctor.

Patient List

Search by patient name...

Scheduled

ID	Patient Name	Symptoms	Appointment Date	Status	Actions
2	pardhiva	Cold	Apr 21, 2025	Scheduled	
6	Mandavalli Pardhiva Sri Sai	Headache	Apr 22, 2025	Scheduled	

By clicking on the “**Actions**” option, the Doctor can access the patient's previous test results and treatment records.

Patient ID: 6

Previous Tests

Blood Test - AIDS +ve,Low Calcium (4/19/2025)

CT Scan - all are fine.but eat apple every day (4/19/2025)

Hepatitis - liver problem (4/22/2025)

X-Ray - bone collision,more teeth , 2 noses, 2 eyes, 2 ears, 1p,2 legs,2 hands (4/21/2025)

Previous Treatments

4/19/2025: dgngnnf (xrhht)

2. Inbox:

The Doctor can view a detailed list of all their appointments, including the date, time, and priority, by clicking on the “Inbox” option.

Today Appointments
1

Scheduled
1

Completed
3

High Priority
1

Appointment Management

Doctor ID: 1

All Appointments

Refresh

Name: kartheek

Priority: High

Date: 4/21/2025

Symptoms: fever

Prescription

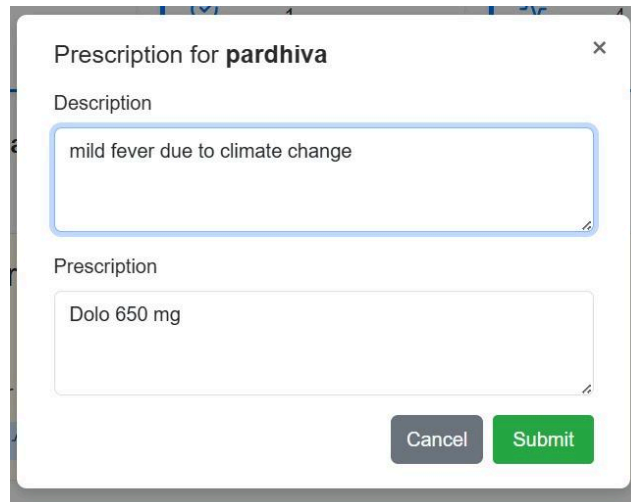
Conduct Test

Cancel

Patient Details

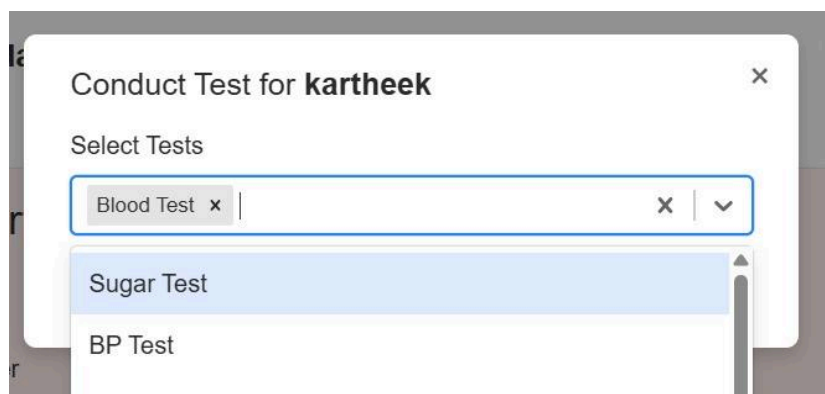
For each appointment, the Doctor has the following options:

- **Prescribe Medicine:** The doctor can prescribe the necessary medications to the patient based on the diagnosis during the appointment.



A screenshot of a web application dialog box titled "Prescription for pardhiva". It contains two text input fields: "Description" with the text "mild fever due to climate change" and "Prescription" with the text "Dolo 650 mg". At the bottom right are "Cancel" and "Submit" buttons.

- **Refer for Tests:** If required, the doctor can refer the patient for specific medical tests, ensuring that the necessary diagnostics are performed.



A screenshot of a web application dialog box titled "Conduct Test for kartheek". It features a "Select Tests" section with a multi-select dropdown menu. The menu is open, showing "Blood Test" (selected), "Sugar Test", and "BP Test".

- **Cancel Appointment:** The doctor can cancel an appointment if it is no longer needed, ensuring the schedule remains updated and accurate.

3. Test Results:

When a doctor refers a patient for tests, the nurse is responsible for conducting the tests and entering the results into the system. These results are then displayed under the **"Test Results"** section, where the doctor can review them to make informed decisions regarding the patient's treatment.

 Patient Test Results Review

Doctor ID: 1

Patients with Completed Tests

Patient ID	Name	Age	Gender	Tests Count	Action
1	kartheek	89	Male	2	<button>Review</button>

4. Post-Test Actions:

Upon receiving the test results, the doctor can take the following actions:

- **Recommend Admission:** If necessary, the doctor can recommend that the patient be admitted to a room for further observation or treatment.
- **Prescribe Medicine and Discharge:** Alternatively, the doctor can prescribe the necessary medication and discharge the patient, sending them home with appropriate instructions for follow-up care.

 Patient Test Results Review

Doctor ID: 1

kartheek (ID: 1)

Patient Info

Age: 89

Gender: Male

Test Details

Test Name	Date	Result
Blood Test	21 Apr 2025	O +ve , low Red Blood Cells
Sugar Test	21 Apr 2025	high sugar levels

← Back to List

🚑 Doctor's Action – Admit or Discharge

✓ Admit Patient

✗ Discharge

5. Add Tests:

The doctor can add new tests to the system by selecting the “**Add Tests**” option. This action ensures that the tests table remains up-to-date with the latest medical tests.

Add New Test

×

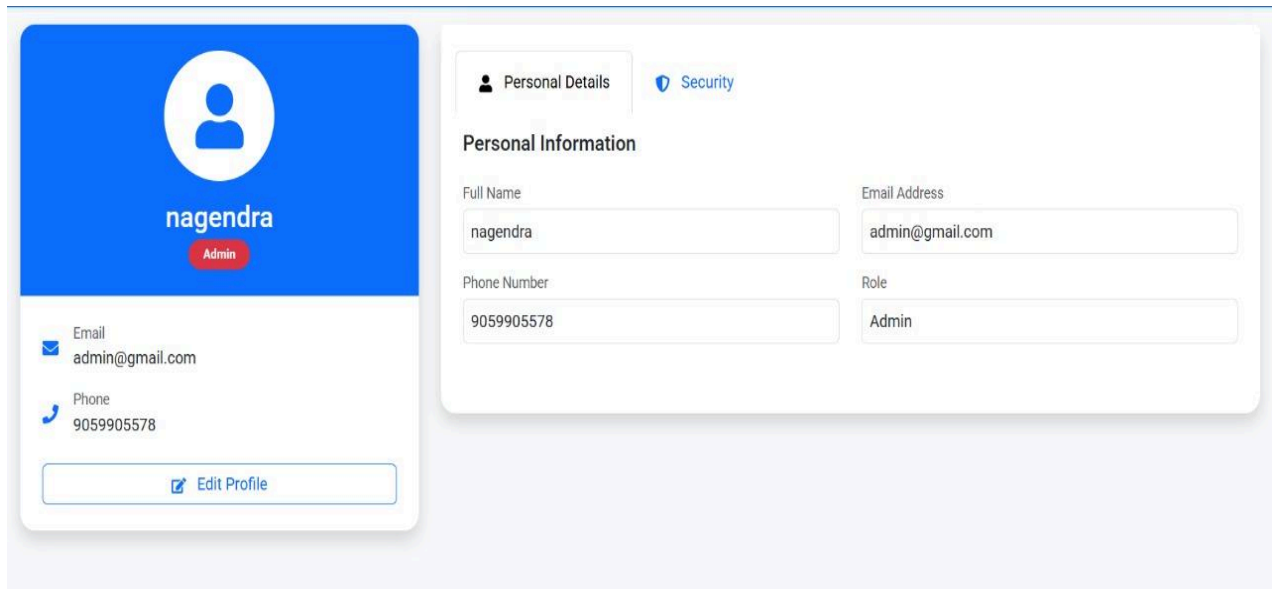
Test Name

Cancel

Add Test

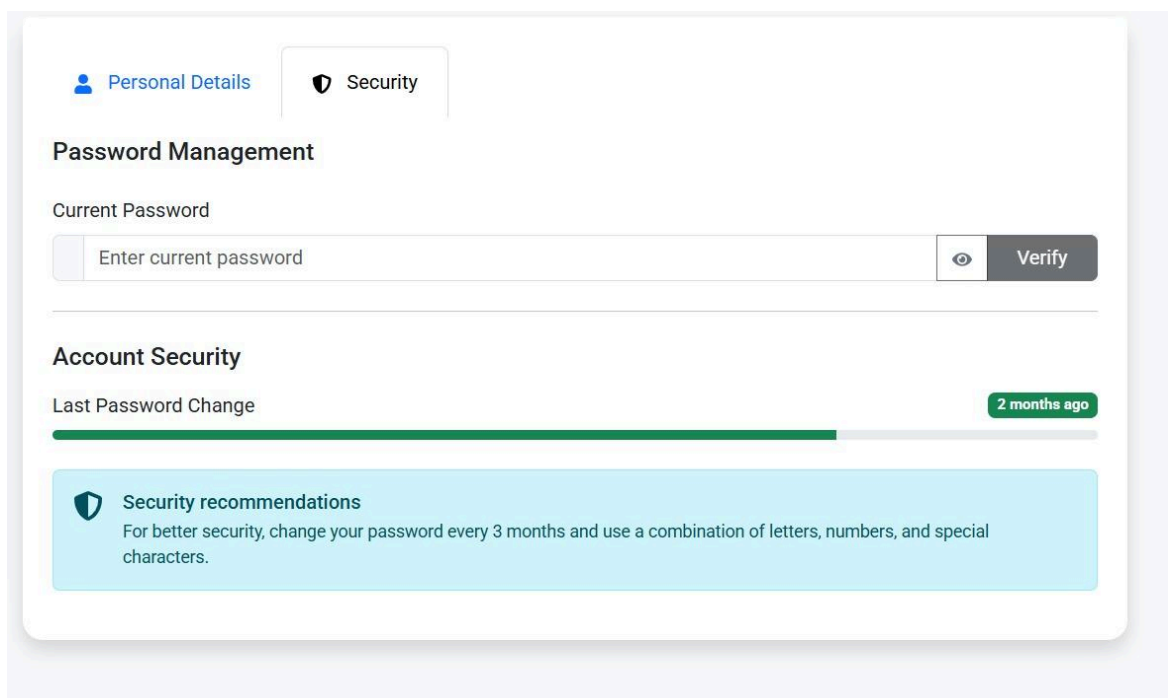
Profile:

The Doctor can access and manage their profile through the “Profile” section. In this section, the doctor can view and edit personal details, such as name, contact information, and professional qualifications. Additionally, the doctor has the ability to update their password for security purposes.



The image shows a user profile interface. On the left is a profile card for 'nagendra' with the role 'Admin'. It displays contact information: Email 'admin@gmail.com' and Phone '9059905578'. An 'Edit Profile' button is at the bottom. To the right is a 'Personal Information' form with two tabs: 'Personal Details' (active) and 'Security'. The form contains four input fields: 'Full Name' (nagendra), 'Email Address' (admin@gmail.com), 'Phone Number' (9059905578), and 'Role' (Admin).

Personal Information	
Full Name	nagendra
Email Address	admin@gmail.com
Phone Number	9059905578
Role	Admin



The image shows the 'Security' tab of the user profile. It features a 'Password Management' section with a 'Current Password' field (placeholder: 'Enter current password') and a 'Verify' button. Below this is an 'Account Security' section showing 'Last Password Change' as '2 months ago' with a progress bar. A 'Security recommendations' box at the bottom advises changing passwords every 3 months and using a combination of letters, numbers, and special characters.

Security

Password Management

Current Password

Enter current password Verify

Account Security

Last Password Change 2 months ago

Security recommendations
For better security, change your password every 3 months and use a combination of letters, numbers, and special characters.

NURSE:

Nurse Dashboard



Patient Data Entry

Enter test results and observations

Enter Data



Document Update

Upload patient-related documents and files

Update Documents

Patient Data Entry:

The Nurse can enter test results for patients who have been prescribed tests by the doctor.

Test Results Entry

Enter test results for pending tests

Patient ▼

kartheek

Doctor: Dr.Pardhiva

Blood Test

Test Conducted Date:

21-04-2025

Test Result:

O +ve , low Red Blood Cells

✓ Submit Results

Future Scope:

In future versions of the system, the **Nurse** will be able to **update/upload test results documents**. Currently, this functionality has not been implemented, but it is planned for inclusion in an improved version of the system.

Summary:

- Patient Registration: Patients register themselves through the Frontdesk Operators.
- Appointment Booking: Frontdesk Operators also assist in booking appointments for the patients.
- Doctor's Role:
 - Doctors view the scheduled appointments.
 - Doctors treat the patients and decide whether to send them for tests or prescribe medicine and discharge the patient.
- Test Results:
 - The Nurse conducts the required tests and uploads the test results.
 - The test results are displayed in the Doctor's Panel under Test Results.
- Doctor's Actions Post-Tests:
 - After reviewing the test results, the doctor can either admit the patient to a room or prescribe medicine and discharge them.
- Room Assignment:
 - If the patient is admitted, the Frontdesk Operator assigns the patient an available room.
- Discharge:
 - After treatment is complete, the Frontdesk Operator discharges the patient from the room, making it available for other patients

Contributions:

- **K. Nagendra (22CS01026):** Designed and developed the Frontend and User Interface for the Doctor and Admin modules.
- **M. Pardhiva Sri Sai (22CS01003):** Contributed to the Frontend development, focusing on the Front Desk and Nurse modules.
- **R. Kartheek (22CS01012):** Handled Backend Development, including API design, database integration, and implementation of logic.
- **V. Aditya Royal (22CS01028):** Designed and implemented the Database Schema, ensuring efficient data organization and relationships across the system.